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Oefeningenreeks 1C nr. 27.5

$$\begin{aligned}(2 \operatorname{cis} 10^\circ)^4 (\operatorname{cis} 16^\circ)^5 &= 2^4 \operatorname{cis}(4 \cdot 10^\circ) \operatorname{cis}(5 \cdot 16^\circ) \\&= 16 \operatorname{cis} 40^\circ \operatorname{cis} 80^\circ \\&= 16 \operatorname{cis}(40^\circ + 80^\circ) \\&= 16 \operatorname{cis} 120^\circ\end{aligned}$$

$$\begin{aligned}16 \operatorname{cis} 120^\circ &= 16 (\cos 120^\circ + i \sin 120^\circ) \\&= 16 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\&= -8 + 8\sqrt{3}i\end{aligned}$$

References